REQUIREMENT ANALYSIS • VERSION 1.0 • FINAL

# AI-Powered IT Service Desk & Incident Management System

*Understanding what needs to be built, before deciding how to build it.*

| Field | Detail |
|---|---|
| **Program** | MP Online Internship Program |
| **Project Type** | C# / ASP.NET Core |
| **Document** | Requirement Analysis |
| **Team Size** | 7 Members |
| **Date** | 13 August 2026 |
| **Version** | 1.0 — Final |

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# 1. Position of Requirement Analysis in the Project Lifecycle

Requirement Analysis is the first major activity in the project planning and design phase. It establishes the foundation for system design, database design, Agile sprint planning, and development. Every later stage of this project depends on the requirements captured here being accurate and complete.

| | | | | | |
|---|---|---|---|---|---|
| **Business Problem** | **Requirement Analysis** | **System Design** | **Database Design** | **Agile Sprint Planning** | **Development** |

**KEY PRINCIPLE**

*Good software development starts with understanding the problem clearly before deciding on the technology or writing a single line of code.*

The purpose of this Requirement Analysis document is to establish a clear, shared understanding of the business problem, the users, the required functionality, the applicable business rules, the security and performance expectations, and the conditions under which a feature will be considered complete — before development begins.

# 2. What Is Requirement Analysis?

Requirement Analysis is the process of identifying, understanding, documenting, and validating what an organization needs from a software system. It helps the development team understand the expected functionality, the users and their roles, the applicable business rules, the security and performance expectations, and the conditions for accepting the finished system.

For this project, Requirement Analysis defines what an organization needs from a centralized platform to manage IT incidents and service requests, with AI assistance built in. The analysis identifies:

- Business problems that need to be addressed
- Stakeholders affected by the system
- Users and their responsibilities
- Functional requirements
- Non-functional requirements
- AI-assisted requirements
- Automation requirements
- User stories and acceptance criteria
- Business rules and conditions for acceptance

**FOCUS OF THIS STAGE**

*Understanding WHAT the organization needs — not deciding how the system will technically be implemented. Technology choices such as ASP.NET Core and SQL Server are decided later, during System Design.*

## 3. Why Requirement Analysis Is Important

Consider a typical customer statement: "We need an IT help desk application where employees can report problems and the support team can resolve them." This is a useful starting point, but it is not detailed enough for development.

The team still needs answers to questions such as: Who can create an incident? What information is required? Who can assign incidents? What priorities and statuses are available? Should notifications be sent automatically? What can administrators manage? Requirement Analysis turns these open questions into clear, documented requirements.

In practice, organizations commonly receive IT support requests through multiple channels — email, phone calls, messaging platforms, or informal conversation. Without a centralized system, this creates several recurring problems:

- IT requests are difficult to track and may be lost or overlooked.
- Multiple employees may report the same problem separately.
- Support teams struggle to determine priority and ownership.
- Incidents are assigned manually, which does not scale.
- Employees have no visibility into the status of their requests.
- Support teams spend significant time searching for past solutions.
- Management has limited visibility into support performance.
- Critical incidents are not escalated quickly enough.
- Historical incident data is difficult to analyze for recurring patterns.

The proposed system addresses these problems with a centralized IT Service Desk platform that additionally provides AI-assisted capabilities:

- Incident classification
- Priority / severity suggestions
- Suggested resolutions
- Duplicate / related-incident detection
- Conversation summarization
- AI-based support assistance

**IMPORTANT**

*AI is designed to assist support staff, not replace their judgment.*

# 4. Case Study — AI-Powered IT Service Desk & Incident Management System

## 4.1 Business Problem

The organization currently manages IT incidents and service requests through disconnected communication channels, with no centralized system of record. As the volume of requests grows, Help Desk Agents face increasing difficulty in tracking incidents, assigning them fairly, prioritizing correctly, communicating with employees, finding relevant past resolutions, monitoring unresolved work, detecting recurring or related incidents, escalating critical issues on time, and measuring support performance. Employees, in turn, have no single place to submit a request and follow its progress.

## 4.2 Proposed Solution

The proposed system is a centralized, AI-powered IT Service Desk and Incident Management platform. Employees report incidents and service requests through the platform and track them end-to-end. Help Desk Agents view, assign, update, and resolve incidents, and communicate directly with employees within the same system. Built-in AI assistance analyzes incoming incidents to suggest categories, priority levels, and possible resolutions, and to flag related or duplicate incidents — while automated workflows handle notifications and escalations so that nothing falls through the cracks.

## 4.3 Business Goal

**BUSINESS GOAL**

*Build a centralized IT Service Desk platform that allows employees to report IT incidents and service requests, while enabling support teams to efficiently classify, prioritize, assign, track, resolve, and analyze those requests with AI-assisted support and automated workflows.*

**SCOPE NOTE**

*At this stage, the focus stays on understanding the business problem — not on choosing ASP.NET Core, SQL Server, or any other implementation technology. Technology decisions belong to the System Design stage.*

## 4.4 High-Level System Concept

**Employees**

*Create and track incidents & service requests*

↓

**AI-POWERED IT SERVICE DESK PLATFORM**

| Incident Management | AI Analysis & Assistance | Notification & Automation |
|---|---|---|

↓

**Help Desk / IT Support Team**

*Review, assign, and resolve incidents*

↓

**Resolution & Closure**

*Recorded, notified, and available in history*

## 5. Stakeholders

A stakeholder is a person, group, or organization that has an interest in the system, or is affected by it, even if they do not use it directly.

| Stakeholder | Interest / Need |
|---|---|
| **Employees** | Report IT problems and service requests, and track their progress. |
| **Help Desk Agents** | View, analyze, assign, update, and resolve incidents efficiently. |
| **IT Administrators** | Manage users, roles, incident categories, and system configuration. |
| **IT Managers** | Monitor support activity, escalations, workload, and performance reports. |
| **Organization Management** | Obtain visibility into overall IT support performance and recurring problems. |

## 6. Actors

An actor is a user or external system that directly interacts with the application through its interface.

| Actor | Description |
|---|---|
| **Employee** | Reports incidents and service requests, and tracks their status. |
| **Help Desk Agent** | Manages assigned incidents and performs day-to-day support activities. |
| **IT Administrator** | Manages users, roles, incident categories, and system |

| Actor | Description |
|---|---|
| | configuration. |
| **IT Manager** | Monitors incidents, escalations, support performance, and reports. |

**IMPORTANT DISTINCTION**

*The AI analysis engine and the database are part of the system's internal infrastructure. They support the actors above but are not treated as independent user actors, since they do not initiate action on their own — they respond to and support the four human roles listed here.*

## 7. Functional Requirements

A functional requirement describes what the system should do. The table below gives a quick summary by actor; detailed, numbered requirements follow in the subsections.

| Actor | Functional Requirements (Summary) |
|---|---|
| **Employee** | Create incidents; view own incidents; track status; communicate with agents. |
| **Help Desk Agent** | View and manage assigned incidents; update status; assign/reassign; resolve. |
| **IT Administrator** | Manage users, roles, categories, and system configuration. |
| **IT Manager** | View reports; monitor unresolved and escalated incidents; track performance. |

### 7.1 Employee Requirements

| ID | Requirement |
|---|---|
| **FR-01** | The employee shall be able to create an IT incident or service request. |
| **FR-02** | The employee shall be able to provide an incident title and description. |
| **FR-03** | The employee shall be able to attach relevant supporting information to an incident. |
| **FR-04** | The employee shall be able to view all incidents they have submitted. |
| **FR-05** | The employee shall be able to track the current status of an incident. |

| ID | Requirement |
|---|---|
| **FR-06** | The employee shall be able to communicate with the assigned support agent. |
| **FR-07** | The employee shall receive notifications about important incident updates. |
| **FR-08** | The employee shall be able to provide feedback after an incident is resolved. |

## 7.2 Help Desk Agent Requirements

| ID | Requirement |
|---|---|
| **FR-09** | The Help Desk Agent shall be able to view incidents assigned to them. |
| **FR-10** | The Help Desk Agent shall be able to update the status of an incident. |
| **FR-11** | The Help Desk Agent shall be able to assign or reassign incidents where authorized. |
| **FR-12** | The Help Desk Agent shall be able to modify incident priority where authorized. |
| **FR-13** | The Help Desk Agent shall be able to communicate directly with employees. |
| **FR-14** | The Help Desk Agent shall be able to record investigation and resolution details. |
| **FR-15** | The Help Desk Agent shall be able to mark an incident as resolved. |
| **FR-16** | The Help Desk Agent shall be able to close a completed incident. |
| **FR-17** | The Help Desk Agent shall be able to view the full history of an incident. |

## 7.3 Administrator Requirements

| ID | Requirement |
|---|---|
| **FR-18** | The Administrator shall be able to create, update, and deactivate user accounts. |
| **FR-19** | The Administrator shall be able to manage user roles and permissions. |
| **FR-20** | The Administrator shall be able to manage incident categories. |

| ID | Requirement |
|---|---|
| **FR-21** | The Administrator shall be able to manage system configuration settings. |
| **FR-22** | The Administrator shall be able to view relevant system activity logs. |

## 7.4 IT Manager Requirements

| ID | Requirement |
|---|---|
| **FR-23** | The IT Manager shall be able to view overall incident statistics. |
| **FR-24** | The IT Manager shall be able to monitor unresolved incidents. |
| **FR-25** | The IT Manager shall be able to monitor escalated incidents. |
| **FR-26** | The IT Manager shall be able to monitor support-team performance. |
| **FR-27** | The IT Manager shall be able to view incident and support reports. |
| **FR-28** | The IT Manager shall be able to identify recurring incident patterns. |

## 7.5 AI-Assisted Functional Requirements

AI assistance is a defining feature of this platform. It supports Help Desk Agents during incident processing but does not replace their final decisions.

| ID | Requirement |
|---|---|
| **AI-01** | The system shall analyze newly submitted incident descriptions. |
| **AI-02** | The system shall suggest an appropriate incident category. |
| **AI-03** | The system shall suggest an appropriate priority or severity level. |
| **AI-04** | The system shall suggest possible resolutions based on available support information. |
| **AI-05** | The system shall identify potentially related or duplicate incidents. |
| **AI-06** | The system shall generate summaries of lengthy incident conversations. |

| ID | Requirement |
|---|---|
| **AI-07** | The system shall provide an AI-based assistant for common IT support queries. |
| **AI-08** | Authorized support personnel shall be able to accept an AI recommendation. |
| **AI-09** | Authorized support personnel shall be able to override an AI recommendation. |
| **AI-10** | AI recommendations shall be presented as assistance and shall never automatically override a human decision. |

## AI Processing Flow

| Incident Submitted | AI Analysis | Category / Priority Suggestion | Resolution / Duplicate Check | Agent Review | Accept or Override |
|---|---|---|---|---|---|

## 7.6 Automation Requirements

The system shall support automated workflows for important incident events, reducing manual follow-up work.

| ID | Requirement |
|---|---|
| **AR-01** | The system shall generate notifications when important incident events occur. |
| **AR-02** | The system shall notify relevant users when an incident is assigned. |
| **AR-03** | The system shall notify employees when their incident status changes. |
| **AR-04** | The system shall support automated escalation when defined conditions are met. |
| **AR-05** | The system shall notify appropriate personnel about critical incidents. |
| **AR-06** | The system shall maintain a record of important automated actions taken. |

*The specific automation technology will be determined during the System Design stage.*

# 8. Non-Functional Requirements

A non-functional requirement describes how the system should behave, or the constraints under which it should operate, rather than a specific feature. Examples of non-functional concerns for this platform include response time, security, availability, scalability, reliability, and maintainability.

## 8.1 Performance

| ID | Requirement |
|---|---|
| **NFR-01** | The system should provide normal application responses within an acceptable time under expected usage. |
| **NFR-02** | Incident searches should return results efficiently, even as incident volume grows. |
| **NFR-03** | Dashboard information should load within an acceptable response time. |
| **NFR-04** | AI processing should provide feedback without unnecessarily blocking the user's workflow. |

## 8.2 Security

| ID | Requirement |
|---|---|
| **NFR-05** | Only authenticated users should be able to access protected functionality. |
| **NFR-06** | Users should only be able to access functionality permitted by their assigned role. |
| **NFR-07** | Administrative functionality should be restricted to authorized administrators only. |
| **NFR-08** | Sensitive incident and user information should be protected at all times. |
| **NFR-09** | Important user and system actions should be auditable. |

## 8.3 Availability

| ID | Requirement |
|---|---|
| **NFR-10** | The system should remain available during normal organizational operating hours. |
| **NFR-11** | Failure of an individual automation workflow should not prevent access to core incident-management functionality. |

### 8.4 Scalability

| ID | Requirement |
|---|---|
| **NFR-12** | The system should support growth in the number of users and incidents over time. |
| **NFR-13** | The system should allow additional AI capabilities to be introduced in the future. |
| **NFR-14** | The system should allow additional integrations and automated workflows to be added. |

### 8.5 Reliability

| ID | Requirement |
|---|---|
| **NFR-15** | Incident information should not be lost during normal system operation. |
| **NFR-16** | Incident status changes should be recorded consistently and accurately. |
| **NFR-17** | Important system actions should maintain an appropriate audit history. |

### 8.6 Maintainability

| ID | Requirement |
|---|---|
| **NFR-18** | The system should be organized into maintainable, well-separated functional modules. |
| **NFR-19** | Business rules should be maintainable without unnecessarily affecting unrelated functionality. |
| **NFR-20** | The system should support future enhancement of AI and automation functionality. |

## 9. Functional vs Non-Functional Requirements

It is useful to keep the distinction between the two categories clear while reviewing requirements:

| Functional — What? | Non-Functional — How well / under what constraints? |
|---|---|
| **Create an incident** | Response time |
| **Assign an incident** | Security |
| **Change incident status** | Performance |

| Functional — What? | Non-Functional — How well / under what constraints? |
|---|---|
| **Generate a support report** | Availability |
| **Suggest an AI category / priority** | Reliability |
| **Send a notification** | Scalability |

## 10. Quick Classification Examples

A quick way to sanity-check a requirement is to ask whether it describes a feature (functional) or a quality/constraint (non-functional):

| Statement | Classification |
|---|---|
| **“The employee should be able to create an incident.”** | Functional |
| **“The application should respond within 2 seconds.”** | Non-Functional |
| **“The administrator should be able to deactivate a user.”** | Functional |
| **“Only authorized administrators should be able to manage users.”** | Non-Functional |
| **“The AI engine should suggest a category for a new incident.”** | Functional |
| **“AI suggestions should never automatically override a human decision.”** | Non-Functional |

## 11. User Stories

In Agile development, requirements are often represented as user stories. A common format is used throughout:

**FORMAT**

*As a [user], I want [functionality], so that [benefit].*

### Employee

- US-01 — As an employee, I want to create an IT incident so that I can report a technical problem to the support team.
- US-02 — As an employee, I want to track my incident status so that I know the progress of my request.
- US-03 — As an employee, I want to receive notifications about my incident so that I know when action has been taken.

- US-04 — As an employee, I want to communicate with the support agent so that I can provide additional information about my problem.

### Help Desk Agent

- US-05 — As a Help Desk Agent, I want to view and manage assigned incidents so that I can take responsibility for resolving them.
- US-06 — As a Help Desk Agent, I want AI-generated category and priority suggestions so that I can process incidents more efficiently.
- US-07 — As a Help Desk Agent, I want suggested resolutions so that I can identify possible solutions quickly.
- US-08 — As a Help Desk Agent, I want to identify related incidents so that recurring problems can be investigated more effectively.
- US-09 — As a Help Desk Agent, I want to summarize lengthy incident conversations so that I can understand the issue quickly.

### Administrator

- US-10 — As an Administrator, I want to manage users and roles so that I can control access to the system.
- US-11 — As an Administrator, I want to manage incident categories so that incidents can be organized consistently.

### IT Manager

- US-12 — As an IT Manager, I want to view incident reports so that I can monitor support activity.
- US-13 — As an IT Manager, I want to monitor unresolved and escalated incidents so that critical problems receive appropriate attention.

## 12. Acceptance Criteria

Acceptance criteria define the conditions that must be satisfied for a user story to be considered complete and acceptable.

**EXAMPLE USER STORY (US-01)**

*As an employee, I want to create an IT incident so that I can report a technical problem to the support team.*

| # | Acceptance Criterion |
|---|---|
| 1 | The employee must be logged into the system. |
| 2 | The employee must provide a valid incident title. |
| 3 | The employee must provide a valid incident description. |
| 4 | The system should create the incident once valid information is submitted. |
| 5 | The system should generate a unique incident number. |

| # | Acceptance Criterion |
|---|---|
| 6 | The initial incident status should be Open. |
| 7 | The employee should be able to view the newly created incident. |
| 8 | The incident should become available for appropriate support-team processing. |

## 13. Given – When – Then

Acceptance criteria can also be expressed using the Given–When–Then structure, which is especially useful for writing test cases directly from requirements.

### Scenario 1 — Create Incident

| | |
|---|---|
| GIVEN | The employee is logged into the system. |
| WHEN | The employee enters a valid incident title and description and submits the incident. |
| THEN | • The system should create the incident.<br>• The system should generate a unique incident number.<br>• The initial status should be Open.<br>• The employee should be able to view the created incident. |

### Scenario 2 — AI Incident Analysis

| | |
|---|---|
| GIVEN | A valid incident has been submitted. |
| WHEN | The system analyzes the incident description. |
| THEN | • The system should generate a suggested category.<br>• The system should generate a suggested priority/severity.<br>• The system may generate a suggested resolution.<br>• Recommendations are presented to an authorized support user, who can accept or override them. |

### Scenario 3 — Incident Assignment

| | |
|---|---|
| GIVEN | An incident has been created and is available to the |

| | |
|---|---|
| | support team. |
| **WHEN** | A Help Desk Agent assigns the incident to themselves or another authorized agent. |
| **THEN** | • The incident should record the assigned agent.<br>• The incident should reflect the appropriate assignment status.<br>• Relevant users should receive an appropriate notification. |

### Scenario 4 — Incident Resolution

| | |
|---|---|
| **GIVEN** | A Help Desk Agent is working on an assigned incident. |
| **WHEN** | The agent records a valid resolution and marks the incident as resolved. |
| **THEN** | • The incident should be marked as Resolved.<br>• The resolution information should be stored.<br>• The employee should be notified.<br>• The incident should remain available in the incident history. |

## 14. From Requirement to Development

Requirement Analysis creates the foundation for the work that follows. Each requirement should remain traceable through system design, database design, and development tasks, all the way to a scheduled sprint.

| | | | | | | |
|---|---|---|---|---|---|---|
| **Business Requirement** | **User Story** | **Acceptance Criteria** | **System Design** | **Database Design** | **Development Task** | **Sprint** |

| Stage | Example for Our Project |
|---|---|
| **Business Requirement** | Employees need a centralized method for reporting IT problems. |
| **User Story** | As an employee, I want to create an IT incident so that I can report a technical problem. |
| **Acceptance Criteria** | Valid incident information creates a unique incident with Open status. |
| **System Design** | Incident interface, controller/API, and application services. |

| Stage | Example for Our Project |
| --- | --- |
| **Database Design** | Incident entity/table and its related data. |
| **Development Task** | Implement incident creation and validation. |
| **Sprint** | Incident Management Sprint. |

*This traceability ensures that every major requirement has a corresponding design and development activity, so nothing agreed upon here is lost by the time the team starts coding.*

## 15. Requirement Analysis Checklist

This checklist was used by the team to confirm the requirement analysis was complete before moving on to System Design.

### Business

☑️ Business problem has been clearly identified.
☑️ Business objective has been defined.
☑️ Stakeholders have been identified.

### Users

☑️ System users have been identified.
☑️ User roles have been defined.
☑️ Responsibilities of each role have been identified.

### Functional

☑️ Incident creation, assignment, status management, and resolution have been defined.
☑️ User management and reporting requirements have been defined.
☑️ AI-assisted requirements have been defined.
☑️ Automation requirements have been defined.

### Non-Functional

☑️ Performance, security, availability, scalability, reliability, and maintainability expectations have all been identified.

### Validation

☑️ Requirements are clear and testable.
☑️ User stories have acceptance criteria.
☑️ Important requirements are traceable to development activities.

## 16. Conclusion & What Comes Next

The proposed AI-Powered IT Service Desk & Incident Management System addresses the problem of fragmented and inefficient IT support management by providing a centralized platform for reporting, tracking, assigning, resolving, and analyzing IT incidents.

The system combines traditional IT service management functionality with AI-assisted incident analysis and automated workflows. AI assists support personnel with recommendations such as incident classification, priority suggestions, possible resolutions, related-incident identification, and conversation summaries — while final decisions remain with the human support team.

The requirements captured in this document become the input for the next modules of the project. The team will progressively move from what needs to be built to how it will be designed and delivered.

| Requirement Analysis | System Design | Database Design | Agile Sprint Planning | Development |
|---|---|---|---|---|

**NEXT STAGE**

*System Design — where these requirements will be converted into the system architecture, modules, interactions, and workflows.*

## Appendix — Team Details

The following team members from the 2023 batch (MP Online Internship Program) contributed to this Requirement Analysis document.

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